Practical 1: Sliding Window.

Aim:
Write a program to implement flow control at data link layer using Sliding Window Protocol. Simulate the flow of frames from one node to another.

⇒ Create a Sender program with following features:
1) Input window size from the user
2) Input a Text message from the user.
3) na

⇒ Create a receiver file with following features
1) Reader a file called Sender_Buffer
2) Check the frame no.
3) If the frame no. are as expected, write the appropriate ACK no. in the Receiver_Buffer file. Eles write NACK no. in the Receiver_Buffer file.

1). Sender Code

```
import time
import os
def sender():
    window_size = int(input("Enter window size: "))
    text = input("Enter message text: ")
    frames = [[i, char] for i, char in enumerate(text)]
    total_frames = len(frames)
    base = 0
    while base < total_frames:
        current_window = frames[base : base + window_size]
        with open("Sender_Buffer.txt", "w") as f:
            for frame in current_window:
                f.write(f"{frame[0]}, {frame[1]}\n")
    print(f"\n[Sender] Sent frames starting at index {base} : {current_window}")
    print("[Sender] waiting for receiver response...")
    time.sleep(2)
    if os.path.exists("Receiver_Buffer.txt"):
        with open("Receiver_Buffer.txt", "r") as f:
            response = f.read().strip().split(":")
        status, ack_num = response[0], int(response[1])
        print(f"[Sender] Received: {status} : {ack_num}")
```

```
        if status == "ACK":
            base = ack_num
        elif status == "NACK":
            print(f"[Sender] Error detected at frame {ack_num}. Resending...")
        else:
            print("[Sender] Receiver response missing, retrying...")

    print("\n[Sender] All frames sent and acknowledged successfully!")

if __name__ == "__main__":
    sender()
```

2) Receiver code

```
import os
import time

def receiver():
    expected_frame = 0
    print("=== Receiver is Running ===")

    while True:
        if os.path.exists("sender_buffer.txt"):
            with open("sender_buffer.txt", "r") as f:
                lines = f.readlines()
            if not lines:
                time.sleep(1)
                continue
            print("\n[Receiver] Processing received batch...")
            all_correct = True
            for line in lines:
                frame_no, data = line.strip().split(":")
```

```
        frame_no = int(frame_no)
        if frame_no == expected_frame:
            print(f"[Receiver] Received Frame {frame_no}: '{data}'")
            expected_frame += 1
        else:
            print(f"[Received] ERROR: Expected Frame {expected_frame}, but got Frame {frame_no}!")
            all_correct = False
            break

with open("Receiver_Buffer.txt", "w") as f:
    if all_correct:
        f.write(f"ACK: {expected_frame}")
        print(f"[Received] Sent ACK: {expected_frame}")
    else:
        f.write(f"NACK: {expected_frame}")
        print(f"[Receiver] Sent NACK: {expected_frame}")
    os.remove("Sender_Buffer.txt")

time.sleep(1)

if __name__ == "__main__":
    receiver()
```

Result:

Hence the sliding window protocol has been success fully executed.